

Q1. What is the relation $\{(1, 2), (2, 3), (3, 4), (4, 5)\}$?

- (A) Function (B) Relation but not a function
(C) Neither relation nor function (D) Both relation and function
(E) None of the above

Q2. Which test can be used to check if a relation is a function?

- (A) Horizontal line test
(B) Vertical line test
(C) Diagonal line test
(D) None of the above
(E) Both A and B

Q3. Given $f(x) = x^2$, what is the domain of $f(x)$?

- (A) $(-\infty, \infty)$
(B) $(-\infty, 0)$
(C) $(0, \infty)$
(D) $[0, \infty)$
(E) None of the above

Q4. For the relation $\{(1, 2), (2, 2), (3, 2)\}$, what is the range?

- (A) $\{1, 2, 3\}$
(B) $\{2\}$
(C) $\{1, 2, 3, 4\}$
(D) $(-\infty, \infty)$
(E) None of the above

Q5. What is $f(2)$ if $f(x) = 2x + 1$?**Q6.** Given the table

x	y
1	2
2	3
3	4

, is this relation a function?

- (A) Yes
(B) No
(C) Maybe
(D) Cannot determine
(E) None of the above

Q7. What is the domain of the relation $\{(x, y) | x^2 + y^2 = 4\}$?

- (A) $[-2, 2]$
(B) $(-2, 2)$
(C) $(-\infty, \infty)$
(D) $(-\infty, -2] \cup [2, \infty)$
(E) None of the above

Q8. Which of the following is a correct notation for a function?

- (A) $f(x) = 2x$
(B) $f = 2x$
(C) $f(x) = x^2 + 1$
(D) $fx = 2x$
(E) All of the above

Open Ended Questions (FRQ)**Q9.** Determine the domain and range of the function $f(x) = \sqrt{x+1}$. Show your work and explain your reasoning.**Q10.** Consider the relation $\{(x, y) | y = x^2 - 4\}$. Determine whether this relation is a function. If it is, identify its domain and range. If it is not, explain why. Provide a clear explanation and show your work.**Chef's Tips**

- Emphasize the importance of the vertical line test for determining if a relation is a function.
- Make sure students understand how to identify domain and range from various representations of functions.
- Highlight the difference between relations and functions using examples.